Candidate's Name/Index Number……..

THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
LONGIDO DISTRICT
ANNUAL JOINT EXAMINATION

FORM ONE
033 BIOLOGY
MARKING SCHEME NOVEMBER 2024

SECTION A (15 MARKS)

1.

i	ii	iii	iv	v	vi	vii	viii	ix	x
B	C	C	B	D	B	B	B	D	A

2.

i	ii	iii	iv	v
C	H	A	E	G

SECTION B (70 Marks)

Answer ALL questions from this section

3. Characteristics of living things:
   - Nutrition: nutrition involves obtaining and using nutrients (food substances) for energy production and growth and repair. Plants manufacture their own food by using simple inorganic substances while animals feed on already manufactured, complex organic substances.
   - Respiration: respiration involves the systematic breaking down of food substances to produce energy. Aerobic respiration requires oxygen while anaerobic respiration does not require oxygen.
   - Reproduction: reproduction involves the formation of new members of a species from the parents.
   - Irritability or sensitivity: irritability refers to response to stimulus.
   - Movement: movement refers to change in posture or location by an organism. Movement that involves the change in location of the whole body is called locomotion. Mostly animals exhibit locomotion, while plants exhibit growth movements.
   - Growth: growth refers to the increase in the size of organisms.
   - Excretion: excretion is the process of removing metabolic wastes from the body. Metabolic wastes are those resulting from various metabolic activities such as respiration and digestion. (Any five @2 = 10 marks)

4. Features of a good Biology laboratory
   - Should have enough supply of air/must be well ventilated.
   - Should have electricity.
   - Should have enough water supplies.
   - Should have fire extinguishers.
   - Should have preserved specimens.

- Should have models.
- Should have oven for drying samples.
- Should have enough space.
- Should have enough heat sources.
- Should have refrigerator or freezer for storing of specimen.
- Should have non-slippery floor.
- Should have proper lighting. (Any ten @1 = 10 marks)

5. Principles of personal hygiene
   - Eating balanced diet regularly to maintain good health
   - The body should be washed daily and hair combed regularly
   - Teeth should be brushed after every meal
   - The nails should be cut short
   - Wear clean and ironed clothes (@1 = 06 marks)
   - Avoid risk behaviors like smoking cigarettes, drinking alcohol, prostitution and others
   - Wash your hands with soap before and after meals and after visiting a lavatory

   Principles of good manners
   - Respect – an individual should have good respect such as…
   - Discipline – an individual should be well disciplined e.g. avoid using bad language; cover, your mouth when sneezing, yawning or coughing; and portray good postures e.g. sitting properly in class and at home.
   - Good eating habits (table manners) which include the following: Eating at regular time intervals; Avoid talking with food is in the mouth; Do not pile food up on your plate. Serve yourself with a reasonable amount while considering
   - Putting things in proper order: All objects in the surrounding should be kept properly and in a right place e.g.: Books should be kept in shelves. (@1 = 04 marks)

6.

Natural passive immunity

Natural immunity is an immunity, which an individual is born with. The individual receive the immunity at birth and from breast-feeding. (2½ marks)

Natural active immunity

Natural active immunity is an immunity, which develops as a result of exposure to pathogens; example, when a person recovers from chicken pox or red eyes. When an individual is exposed to such infections, his or her body responds by producing antibodies to fight off the infection. The body becomes always ready and effective to fight off any such similar infection later on in one's lifetime. (2½ marks)

Artificial passive immunity

Artificial passive immunity is the type of immunity whereby antibodies from another organisms are introduced in the body of an individual.

(2½ marks)

Artificial active immunity

Artificial active immunity is the type of immunity where by attenuated pathogen (pathogen whose ability to cause infection is weakened) is introduced in the body of an individual to trigger off production of antibodies. (2½ marks)

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7. (a)
   (i) Only the first letter of the generic name is capitalized and then the whole generic name is underlined or italicized; example: Homo. 02 marks
   (ii) The first letter of the species name is never capitalized, and then the whole species name is underlined; example: sapiens. So the scientific name for human is Homo – genus, sapiens – species. 02 marks
   (iii) If the scientific name is written in italics, then it should never be underlined, but other rules remain the same. Example Homo sapiens or *Homo sapiens* 02 marks

   (b)
   Disadvantages of Artificial classification
   (i) It is not accurate enough as it uses a few observable features. 01 marks
   (ii) It is biased; *explain*…. 01 marks
   (iii) Ends up with unrelated organisms being put in the same group or vice versa. 01 marks
   (iv) It may cause confusion, *explain..* 01 marks

8. The scientific process
   a. Problem identification and problem question
      This is the step where a problem is identified and acknowledged; for example, wilting of the maize
      This step involves formulation of a question on the problem. For example, 'was too much fertilizers/ soil acidity responsible for wilting/' ……..……..02 marks
   b. Hypothesis formulation
      This step involves stating a hypothesis; a hypothesis is a tentative explanation (an intelligent guess). For the example, 'the acidity of the soil/quantity of fertilizer applied is responsible for the wilting.' ……....02 marks
   c. Experimentation
      Experiment involves investigations conducted to test the hypothesis, that is, to show whether the hypothesis is correct or incorrect.……..01½ marks
   d. Observation and data collection ……..…..01 mark
      This stage involves the recording of data because of the experiment.
   e. Data interpretation
      This is the process of determining the meaning of the collected data. For example, the above data would suggest that the wilting with different quantity of fertilizer/or after decreasing soil acidity by liming was less… ……..01½ marks
      ……..02 marks
   f. Conclusion
      The conclusion either accepts or rejects the hypothesis. For example, in this case, the hypothesis was 'the acidity of the soil/quantity of fertilizer applied is responsible for the wilting.'
      If the result of the experiment supports this hypothesis, then the conclusion is also; 'the acidity of the soil/quantity of fertilizer applied is responsible for the wilting.'
      If the result of the experiment did not support this hypothesis, then the conclusion would be; 'the acidity of the soil/quantity of fertilizer applied is **not** responsible for the wilting.'

9. Advantages:
   (i) They cause decomposition of matter hence releases nutrients to plants.
   (ii) Some aid in digestion, example the bacteria in the gut of the ruminants.
   (iii) Some produce vitamins K and B_{12} in the human gut.

(iv) Some convert atmospheric nitrogen to nitrates for plant use.
(v) Some are used in the production of antibiotics.
(vi) Some are used in making vaccines.
(vii) Some are used in sewage treatment (They decompose pollutants).
(viii) Some are used in the food industry for cheese production, yogurt and sauerkraut production.
(ix) Some are used in genetic engineering to produce insulin.
(x) Autotrophic bacteria release oxygen in the atmosphere.
(xi) Some are used in biological researches. (Any eight.., 1@ = 8 marks)

Disadvantages:
(i) They cause diseases
(ii) They spoil foods (Any two.., 1@ = 2 marks)

SECTION C (15 Marks)

10. Intro: Tuberculosis (TB) is a bacterial disease that mainly affects the lungs and causes serious illness *(01 mark)*

Cause: TB is caused by a bacterium called *mycobacterium tuberculois*/ caused by bacteria that multiply in the lungs and cause disease there and the infection may spread to other parts of the body outside the lungs extra-pulmonary tuberculosis) *(01 mark)*

Transmission

Tuberculosis is transmitted through air when a sick person coughs, sneeze, sing, etc., which can put tiny droplets with germ in the air. When another person breathes in the contaminated droplets, the germ enter the lungs and may cause a disease in that person. *(04 marks)*

Signs and symptoms:

- Low fever, tiredness and cough
- Coughing up blood or mucus
- Chest pain, painful breathing or coughing *(Any five 01@ =05 marks)*
- Fever and chills
- Night sweat
- Fatigue and loss of appetite
- Abnormal inflammation of lymph nodes.
- Abnormal mental condition signs.

Control and treatment:

- Isolation
- wearing face mask
- Enough ventilation *(Any four 01@ =04 marks)*
- Covering mouth when coughing and sneezing.
- Vaccinantion.
- TB medications

Conclusion: TB leads to a serious illness. Identifying risk factors may be the best way to avoid TB… (or any other relevant conclusion: *01 mark)*

Total: 15 marks